Data Structures and Algorithms - Python Assessment (synthetic sample, harder)

Use only the Python standard library. Do not use input(). Write functions/classes that return results.

Question 1. Write a Python class LRUCache with __init__(self, capacity), get(self, key) and put(self, key, value). get returns the value or -1 if the key is missing. Both operations must be O(1) and a get counts as a use. When the cache is full, put evicts the least recently used key.

Question 2. Write a Python function merge_intervals(intervals) that takes a list of [start, end] pairs (unsorted, possibly overlapping or touching) and returns the merged list sorted by start. Touching intervals such as [1,2] and [2,3] must be merged into [1,3].

Question 3. Write a Python function eval_rpn(tokens) that evaluates a Reverse Polish Notation expression given as a list of strings with the operators + - * /. Division truncates toward zero. Raise ValueError for an invalid expression.

Question 4. Write a Python function shortest_path(graph, start, goal) where graph is a dict mapping a node to a dict of {neighbour: positive weight}. Return a tuple (total_distance, path_as_list_of_nodes) for the cheapest route, or None if the goal cannot be reached.

Question 5. Write a Python function group_anagrams(words) that returns a list of groups (each a sorted list of words) of words that are anagrams of each other, with the groups ordered by their first word.